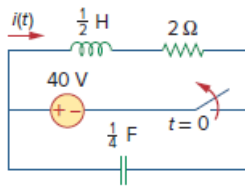


Problem

The switch in the circuit of Fig. 16.47 has been closed for a long time but is opened at $t = 0$. Determine $i(t)$ for $t > 0$.

Figure 16.47:



Step-by-step solution

Step 1 of 7

Refer to Figure 16.47 in the text book.

Draw the circuit diagram at $t < 0$.

At $t < 0$, inductor acts as short circuit and capacitor acts as open circuit.

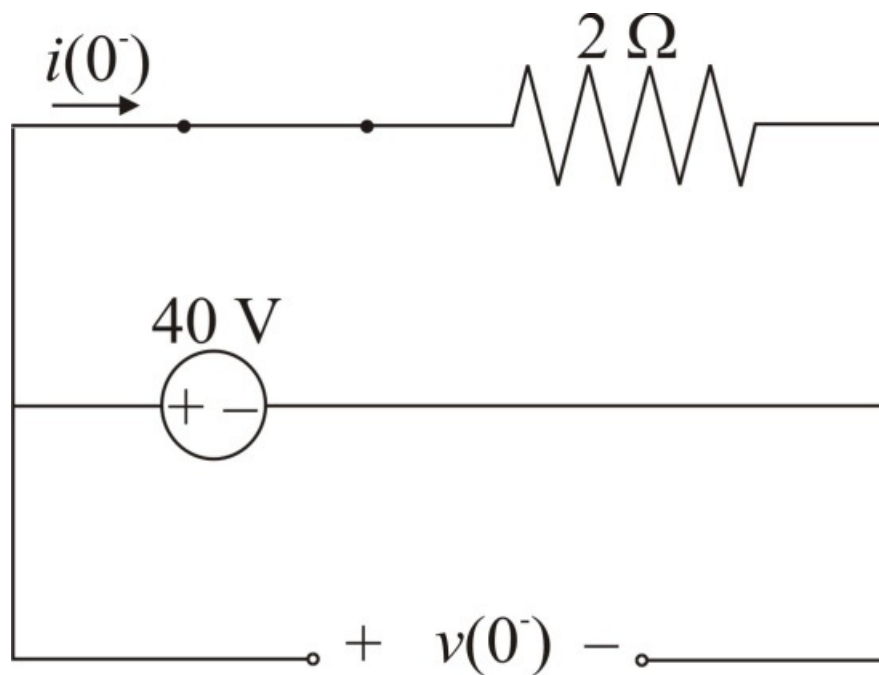


Figure 1

Step 2 of 7

The value of the voltage $v(0^-)$ is 40 V.

Calculate the value of the current $i(0^-)$.

$$\begin{aligned} i(0^-) &= \frac{40}{10} \\ &= 4 \text{ A} \end{aligned}$$

Therefore, the value of the current $i(0^-)$ is 4 A.

[Comments \(2\)](#)



Anonymous

Should be 40 divided by 2.



Anonymous

$i(0)=20 \text{ A}$

Step 3 of 7

Draw the circuit diagram at $t > 0$. Represent the circuit parameters in Laplace transform.

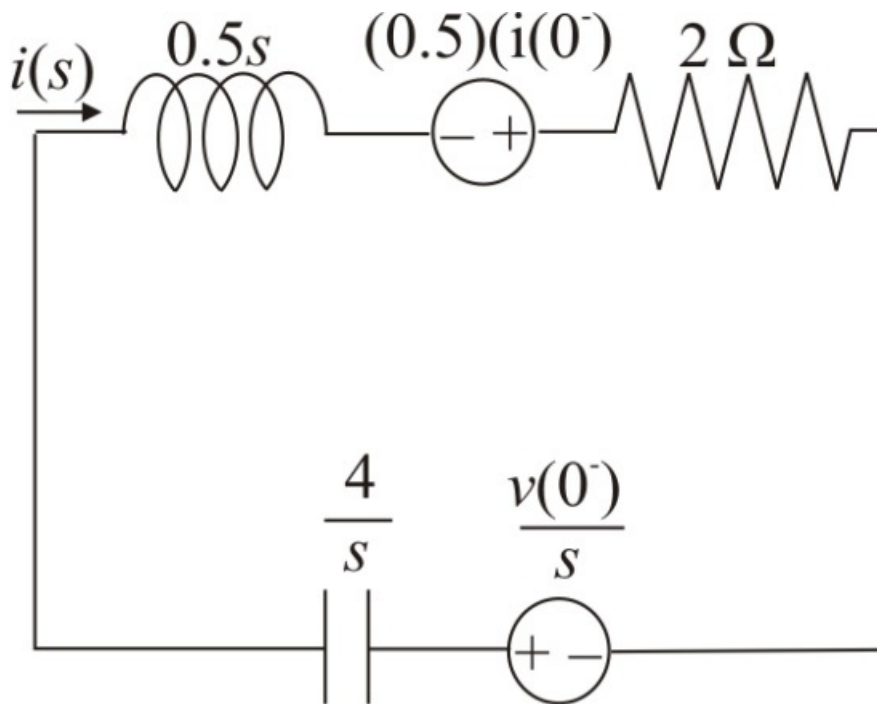


Figure 2

Step 4 of 7

Apply Kirchhoff's Voltage Law in the loop.

$$i(s)\left(0.5s + \frac{4}{s} + 2\right) - 0.5i(0^-) - \frac{v(0^-)}{s} = 0$$

Substitute 40 V for $v(0^-)$ and 20 A for $i(0^-)$.

$$i(s)\left(0.5s + \frac{4}{s} + 2\right) - 0.5(20) - \frac{40}{s} = 0$$

$$i(s)\left(\frac{0.5s^2 + 2s + 4}{s}\right) - 10 - \frac{40}{s} = 0$$

$$i(s)\left(\frac{0.5s^2 + 2s + 4}{s}\right) = \frac{10s + 40}{s}$$

$$i(s)(0.5s^2 + 2s + 4) = 10s + 40$$

$$\begin{aligned} i(s) &= \frac{10s + 40}{0.5s^2 + 2s + 4} \\ &= \frac{10s + 40}{0.5(s^2 + 4s + 8)} \\ &= \frac{20s + 80}{s^2 + 4s + 8} \end{aligned}$$

$$i(s) = \frac{20s + 80}{(s + 2 - j2)(s + 2 + j2)} \dots\dots (1)$$

[Comments \(1\)](#)



Anonymous

Would it not be easier to complete the square for the denominator here?

Step 5 of 7

Consider $\frac{20s + 80}{(s + 2 - j2)(s + 2 + j2)}$ to be $\frac{A_1}{s + 2 + j2} + \frac{A_2}{s + 2 - j2}$.

$$\frac{20s + 80}{(s + 2 - j2)(s + 2 + j2)} = \frac{A_1}{s + 2 + j2} + \frac{A_2}{s + 2 - j2}$$

Apply residue method to find the value of the constant A_1 .

$$\begin{aligned} A_1 &= (s + 2 - j2) \left(\frac{20s + 80}{(s + 2 - j2)(s + 2 + j2)} \right) \Bigg|_{s = -2 + j2} \\ &= \frac{20s + 80}{(s + 2 + j2)} \Bigg|_{s = -2 + j2} \\ &= \frac{20(-2 + j2) + 80}{-2 + j2 + 2 + j2} \\ &= \frac{-40 + j40 + 80}{j4} \\ A_1 &= \frac{40 + j40}{j4} \\ &= 10 - j10 \end{aligned}$$

Therefore, the value of the constant A_1 is $10 - j10$.

Step 6 of 7

Apply residue method to find the value of the constant A_2 .

$$\begin{aligned} A_2 &= (s+2+j2) \left(\frac{20s+80}{(s+2-j2)(s+2+j2)} \right) \bigg|_{s=-2-j2} \\ &= \frac{20s+80}{(s+2-j2)} \bigg|_{s=-2-j2} \\ &= \frac{20(-2-j2)+80}{-2-j2+2-j2} \\ &= \frac{-40-j40+80}{-j4} \\ A_2 &= \frac{40-j40}{-j4} \\ &= 10+j10 \end{aligned}$$

Therefore, the value of the constant A_2 is $10+j10$.

Step 7 of 7

Recall equation (1).

$$i(s) = \frac{20s + 80}{(s + 2 - j2)(s + 2 + j2)}$$

Substitute $\frac{A_1}{s + 2 + j2} + \frac{A_2}{s + 2 - j2}$ for $\frac{20s + 80}{(s + 2 - j2)(s + 2 + j2)}$.

$$i(s) = \frac{A_1}{s + 2 + j2} + \frac{A_2}{s + 2 - j2}$$

Substitute $10 + j10$ for A_2 and $10 - j10$ for A_1 .

$$\begin{aligned} i(s) &= \frac{10 - j10}{s + 2 + j2} + \frac{10 + j10}{s + 2 - j2} \\ &= 10 \left(\frac{1}{s + 2 + j2} + \frac{1}{s + 2 - j2} \right) + j10 \left(\frac{-1}{s + 2 + j2} + \frac{1}{s + 2 - j2} \right) \end{aligned}$$

Apply inverse Laplace transform.

$$\begin{aligned} i(t) &= 10 \left(e^{-(2-j2)t} + e^{-(2+j2)t} \right) + j10 \left(-e^{-(2-j2)t} + e^{-(2+j2)t} \right) \\ &= 10 \left(e^{(2+j2)t} + e^{(2-j2)t} \right) + j10 \left(-e^{(2+j2)t} + e^{(2-j2)t} \right) \\ &= 10e^{2t} \left(e^{j2t} + e^{-j2t} \right) + j10e^{2t} \left(-e^{j2t} + e^{-j2t} \right) \\ &= 10e^{2t} \left(e^{j2t} + e^{-j2t} \right) - j10e^{2t} \left(e^{j2t} - e^{-j2t} \right) \end{aligned}$$

$$\begin{aligned} i(t) &= 20e^{2t} \left(\frac{e^{j2t} + e^{-j2t}}{2} \right) + 20e^{2t} \left(\frac{e^{j2t} - e^{-j2t}}{2j} \right) \\ &= 20e^{2t} \cos(2t) + 20e^{2t} \sin(2t) \\ &= 20e^{2t} (\cos(2t) + \sin(2t)) \\ &= 20e^{2t} (\cos(2t) - \cos(2t + 90^\circ)) \end{aligned}$$

$$\begin{aligned} i(t) &= 20e^{2t} \left(2 \sin \left(\frac{2t + 2t + 90^\circ}{2} \right) \sin \left(\frac{2t + 90^\circ - 2t}{2} \right) \right) \\ &= 40e^{2t} \sin \left(\frac{4t + 90^\circ}{2} \right) \sin \left(\frac{90^\circ}{2} \right) \\ &= 40e^{2t} \sin(2t + 45^\circ) \sin(45^\circ) \\ &= \frac{40}{\sqrt{2}} e^{2t} \sin(2t + 45^\circ) \end{aligned}$$

$$i(t) = 28.28e^{2t} \sin(2t + 45^\circ) \text{ A}$$

Therefore, the value of the current $i(t)$ is $\boxed{28.28e^{2t} \sin(2t + 45^\circ) \text{ A}}$.

[Comments \(2\)](#)



Anonymous

This answer has a positive exponential....



Anonymous

whoever did this question took the longest possible route, this is a very basic laplace. None of these steps are required.